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Two results on set families: sturdiness and intersection. (English. English summary)

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A family $\mathcal{F} \subseteq 2^{[n]}$ is called an *IU-family* if $\mathcal{F}$ and $\mathcal{F}^c$ are both intersecting families. The classical IU-Theorem states that if $\mathcal{F} \subseteq 2^{[n]}$ is an IU-family, then $|\mathcal{F}| \leq 2^{n-2}$. The notion of *sturdiness* $\beta(\mathcal{F})$ of a family $\mathcal{F}$, introduced by P. Frankl and J. Wang [“The maximum sturdiness of intersecting families”, preprint, arXiv:2412.07090], is defined as $\beta(\mathcal{F}) = \min_{i \neq j} |\mathcal{F}(i, \bar{j})|$, where $\mathcal{F}(i, \bar{j}) := \{F \setminus \{i\} : F \in \mathcal{F}, F \cap \{i, j\} = \{i\}\}$. Frankl and Wang showed that if $\mathcal{F} \subseteq 2^{[n]}$ is an IU-family for which there is a partition $[n] = X \cup Y$ such that both $\mathcal{F}|_X$ and $(\mathcal{F}|_Y)^c$ are intersecting families, then $\beta(\mathcal{F}) \leq 2^{n-4}$. They conjectured that this bound holds for all IU-families $\mathcal{F} \subseteq 2^{[n]}$.

The first result of this paper is a proof of the above conjecture of Frankl and Wang. The proof is quite short and elegant, and it uses a strengthened form of a result of P. D. Seymour [Mathematika **20** (1973), 208–209; MR0347619] on cross-Sperner families, proved via a correlation inequality for down-sets and up-sets due to M. H. Siggers and N. Tokushige [European J. Combin. **33** (2012), no. 2, 128–138; MR2854635] and the AM-GM inequality.

Next, let $[m] = \bigcup_{i \in [k]} X_i$ be a partition of $[m]$ where $|X_i| = n$ for each $1 \leq i \leq k$. Define $\mathcal{H}(n, k, \ell) := \{H \in \binom{[m]}{\ell} : |H \cap X_i| \leq 1 \text{ for all } i \in [k]\}$. A family $\mathcal{F} \subseteq \mathcal{H}(n, k, \ell)$ is said to be *separated*. Frankl et al. [Discrete Math. **346** (2023), no. 12, Paper No. 113574; MR4611141] determined the maximum sum $|\mathcal{F}| + |\mathcal{G}|$ for nonempty cross intersecting separated families $\mathcal{F} \subseteq \mathcal{H}(n, k, \ell)$ and $\mathcal{G} \subseteq \mathcal{H}(n, k, \ell')$ when $n > 3\ell$ and $k \geq \ell \geq \ell' \geq 2$. They also raised the question of whether the same bound holds for all $n \geq 2$ and $k \geq \ell \geq \ell' \geq 1$.

The second result of this paper is a sharp upper bound on $|\mathcal{F}| + |\mathcal{G}|$ for nonempty cross t -intersecting separated families for all $n \geq 2$, thereby generalizing the above result of Frankl, Liu, Wang and Yang as well as a result of E. L. L. Liu [AIMS Math. **8** (2023), no. 12, 30910–30921; MR4670853]. The proof uses the shifting technique due to P. Erdős, C. Ko and R. Rado [Quart. J. Math. Oxford Ser. (2) **12** (1961), 313–320; MR0140419] and the necessary intersection point method of P. Gupta et al. [Bull. Lond. Math. Soc. **55** (2023), no. 3, 1447–1458; MR4618236]. In particular, the authors settle the open problem raised by Frankl, Liu, Wang and Yang in the negative by constructing explicit counterexamples for $t = 1$ and $(n, k, \ell, \ell') = (2, 3, 3, 2)$ and $(2, 4, 4, 2)$.

A minor typo: In the statement of Theorem 1.1, $|\mathcal{F}|$ should read $|\mathcal{G}|$.

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Note: This list reflects references listed in the original paper as accurately as possible with no attempt to correct errors.